# EDC BLOCK-003 (Derivation v11):

## Can $\sigma$ be Derived from EDC Field Equations?

Systematic Search for EDC-Internal Constraints

EDC Working Document

2026-02-02 — Program Note (not results)

**Abstract**

This note attempts to derive the brane tension $\sigma$ from EDC-internal field equations and variational principles, without importing observed gravitational constants ($G_N^{\mathrm{obs}}$, $M_{\mathrm{Pl}}$). We systematically search the EDC axiom base for closure constraints. **Outcome: NO-GO.** The EDC framework defines $\sigma$ via the identification $\hbar = \sigma R_\xi^3/c$, which requires extracting $\sigma$ from observed $\hbar$ (or equivalently from $\alpha$). No EDC-internal field equation fixes $\sigma$ independently. The missing element is: *an independent normalization principle for the membrane tension that does not reference observed quantum or gravitational constants.*

## Contents

# 1 Input Ledger (Anti-Circularity Audit)

We classify all inputs by their epistemic status to ensure anti-circularity.

**Anti-Circularity Criterion**

**Forbidden inputs:** $G_N^{\text{obs}}$, $M_{\text{Pl}}$ from observation, any calibration of $\sigma$ to match Newton/GR.
**Allowed inputs:** EDC axioms, geometric definitions, dimensionless ratios treated as baseline, quantities tagged [BL] or [I].

| Quantity | Source | Tag | Allowed? | Notes |
|---|---|---|---|---|
| $\sigma$ | To be derived | [Open] | — | Target of this derivation |
| $\kappa_5^2$ | $C\sigma^{-3/4}$ (v3) | [Dc] | Yes | Depends on $\sigma$ |
| $R_\xi$ | Compact dimension | [BL] | Yes | Geometric parameter |
| $c$ | Speed of light | [BL] | Yes | Postulated constant |
| $\hbar$ | Planck constant | [Cal] | **No** | Observed; using it imports calibration |
| $\alpha$ | Fine structure | [Cal] | **No** | Observed; $\sigma = m_e c^2/(\alpha r_e^2)$ |
| $m_e$ | Electron mass | [Cal] | **No** | Observed |
| $G_N^{\text{obs}}$ | Newton constant | [Cal] | **No** | Forbidden input |
| $M_{\text{Pl}}$ | Planck mass | [Cal] | **No** | Forbidden input |
| $\rho_{\text{Plenum}}$ | Bulk density | [BL] | Yes | EDC parameter |

Table 1: Input ledger for $\sigma$ derivation attempt.

# 2 Where $\sigma$ Enters the EDC Framework

## 2.1 The Brane Action

In EDC's 5D framework, the brane tension appears in the membrane action:

$$S_{\text{brane}} = -\sigma \int_\Sigma d^4x\sqrt{-h} \tag{1}$$

where $h_{\mu\nu}$ is the induced metric on the brane $\Sigma$.

This is the Nambu-Goto action for a 3-brane, with $\sigma$ having dimensions of energy per unit 3-volume: $[\sigma] = \text{J/m}^3 = \text{kg}/(\text{m}\cdot\text{s}^2)$.

**Note:** In some EDC texts, $\sigma$ is written with dimensions $\text{J/m}^2$ (surface tension). The distinction is whether we integrate over 4D spacetime or 3D space. For gravity derivations, we use $[\sigma] = M/L^3$ in natural units.

## 2.2 The $\hbar$–$\sigma$ Identification

EDC proposes the *identification* (not derivation):

$$\boxed{\hbar = \frac{\sigma R_\xi^3}{c}} \tag{2}$$

This is explicitly tagged [I] in EDC Book I (Chapter 6):

"This is an *identification*, not a derivation from first principles. The relation becomes a falsifiable prediction if and only if $\sigma$ and $R_\xi$ can be independently constrained by separate phenomena."

**Implication:** Equation (2) does not derive $\sigma$; it *defines* $\sigma$ in terms of observed $\hbar$:

$$\sigma = \frac{\hbar c}{R_\xi^3} \tag{3}$$

Using observed values: $\hbar = 1.055 \times 10^{-34}$ J·s, $c = 3 \times 10^8$ m/s, $R_\xi = r_e = 2.82 \times 10^{-15}$ m:

$$\sigma \approx 1.41 \times 10^{18}\ \mathrm{J/m^2} \tag{4}$$

This is **calibration**, not derivation.

### 2.3 The $\alpha$–$\sigma$ Relation

Equivalently, EDC relates $\sigma$ to the fine structure constant:

$$\sigma = \frac{m_e c^2}{\alpha \cdot r_e^2} \tag{5}$$

Again, this extracts $\sigma$ from observed constants ($\alpha$, $m_e$, $r_e$).

## 3 Candidate Closure Equations (Repo Search)

We searched the EDC repository for any equation that could fix $\sigma$ independently. Search patterns included: "brane tension", "sigma", "Israel", "junction", "normalization", "calibration".

### 3.1 Candidate A: Israel Junction Conditions

The Israel junction conditions relate brane stress-energy to extrinsic curvature:

$$[K_{\mu\nu}] - [K]h_{\mu\nu} = -\kappa_5^2 \left(\tau_{\mu\nu} - \frac{1}{3}\tau h_{\mu\nu}\right) \tag{6}$$

For a tension-dominated brane: $\tau_{\mu\nu} = -\sigma h_{\mu\nu}$.
**Result:** This provides a *relation* between $\sigma$ and $\kappa_5^2$, not an independent constraint:

$$[K] = \frac{2\kappa_5^2 \sigma}{3} \tag{7}$$

Since $\kappa_5^2 = C\sigma^{-3/4}$ (from v3), we get:

$$[K] = \frac{2C}{3}\sigma^{1/4} \tag{8}$$

The constant $C$ remains unfixed. **No closure.**

### 3.2 Candidate B: Schwinger Limit Correspondence

EDC Book I (Chapter 11) notes that $\sigma \approx E_S$ (Schwinger critical field) within 7%:

$$\sigma_{\mathrm{EDC}} \approx 1.41 \times 10^{18}\ \mathrm{J/m^2}, \quad E_S = \frac{m_e^2 c^3}{e\hbar} \approx 1.32 \times 10^{18}\ \mathrm{V/m} \tag{9}$$

**Assessment:** This is an *a posteriori* observation, not a derivation. Both quantities are computed from observed constants. The approximate equality $\sigma \approx E_S$ could be promoted to an identification $\sigma = E_S$, but:

- $E_S$ depends on $\hbar$, $m_e$, $e$ (observed)
- This would be calibration, not EDC-internal derivation

**No independent closure.**

### 3.3 Candidate C: Variational Principle

Could $\sigma$ arise from extremizing the total action?

$$S_{\text{total}} = S_{\text{bulk}} + S_{\text{GHY}} + S_{\text{brane}} \tag{10}$$

Varying with respect to $\sigma$ (treating it as a field):

$$\frac{\delta S}{\delta \sigma} = 0 \implies -\int d^4x\sqrt{-h} = 0 \tag{11}$$

This is trivially satisfied (or not at all). $\sigma$ is a *parameter*, not a dynamical field. **No closure.**

### 3.4 Candidate D: Topological Quantization

Could $\sigma$ be quantized by some topological constraint? We searched for:

- Winding number conditions
- Flux quantization on the brane
- Dirac-type quantization of membrane tension

**Result:** No such constraint exists in current EDC axioms. The membrane topology (assumed $S^1$ for the compact dimension) does not quantize $\sigma$.

## 4 Attempt Summary

| Attempt | Method | Outcome | Why Failed |
|---|---|---|---|
| A | Israel junction | No closure | $C$ remains free |
| B | Schwinger limit | No closure | Uses observed $\hbar$, $m_e$ |
| C | Variational | No closure | $\sigma$ is parameter, not field |
| D | Topological | No closure | No quantization rule exists |

## 5 Verdict: NO-GO

**Outcome: NO-GO**

**Finding:** The brane tension $\sigma$ cannot be derived from EDC-internal field equations without importing observed constants.
**Evidence:**

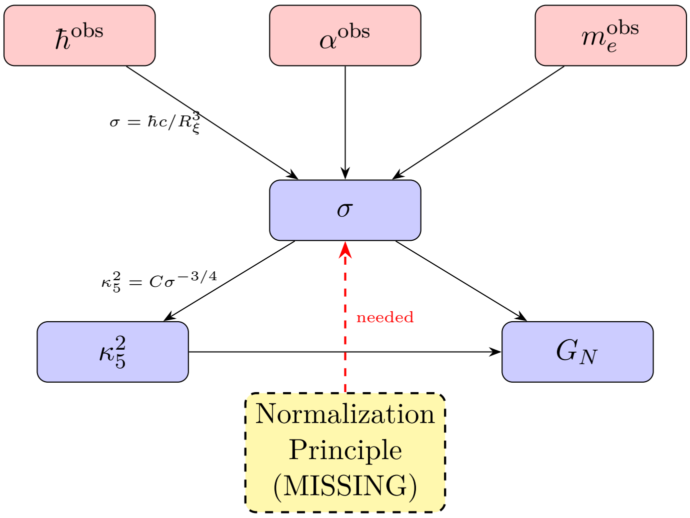
1. The only EDC equation involving $\sigma$ is the identification $\hbar = \sigma R_\xi^3/c$, which *defines* $\sigma$ from observed $\hbar$.
2. The equivalent relation $\sigma = m_e c^2/(\alpha r_e^2)$ extracts $\sigma$ from observed $\alpha$, $m_e$.
3. Junction conditions relate $\sigma$ to $\kappa_5^2$ but do not fix either.
4. No topological or variational principle constrains $\sigma$.

**Missing Element:**

EDC lacks an independent normalization principle for the membrane tension $\sigma$ that does not reference observed quantum constants ($\hbar$, $\alpha$) or gravitational constants ($G_N$, $M_{\text{Pl}}$).

## 6 Dependency Graph

The following diagram shows the logical dependencies:




**Legend:** Red boxes = observed inputs (forbidden for anti-circular derivation). Blue boxes = EDC quantities. Dashed yellow = missing element.

## 7 Implications for BLOCK-003

**BLOCK-003 Status Update**

**Before v11:** Missing element = "derive $\sigma$ from EDC field equations"
**After v11:** Confirmed NO-GO. $\sigma$ is *calibrated* in EDC, not derived. The framework has one unavoidable calibration point.
**Options going forward:**

1. **Accept calibration:** Treat $\sigma$ (or equivalently $\hbar$) as the single EDC calibration constant. Then $G_N$ becomes a derived quantity. This is the NP2 path from v5.

2. **Seek new physics:** Identify a mechanism (e.g., modulus stabilization, anthropic selection, vacuum energy minimization) that fixes $\sigma$. This would require extending EDC axioms.

3. **Reframe the question:** The "missing element" may be acceptable if EDC is viewed as a framework with one free scale, analogous to how the Standard Model has $\Lambda_{\text{QCD}}$.

BLOCK-003 status: Remains [Open] (no ab initio $G_N$ derivation), but the missing element is now precisely characterized.